



Investigation report

Grounding on Boolarong Bank, Weerona Channel, Kanooka Heads -
NTSB-style probable cause report

Drafted 04/10/2026

INVESTIGATION WORKFLOW SUITE

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Executive summary

What happened

On 07/07/2026 MV Tarrakin Star, a 24.4 m aluminium catamaran passenger ferry, was inbound through the Weerona Channel at Kanooka Heads on service TF18. It was dark and the tide was falling. On board were 38 passengers, a master, a deckhand and a kiosk attendant.

At 18:07:06 the chart plotter display froze and did not recover. The vessel continued at about eight knots. Its heading increased from 195 to 200 degrees and held, and it left the dredged channel at about 18:09:00. The echo sounder shallow alarm sounded at 18:11:48. At 18:12:04 the vessel grounded on Boolarong Bank at 6.8 kt, 74 m west of the charted centreline and 54 m outside the channel.

The starboard hull bottom plating was set up over 4.1 m with a 180 mm split, and the No. 2 starboard void flooded to approximately 260 mm. The starboard propeller, shaft and rudder stock were damaged and the skeg was torn away. There were no serious injuries. One passenger reported a grazed shin and two were treated at the wharf for distress. All passengers were ashore by 19:26. The master stopped both engines and did not attempt to refloat the holed vessel, which very likely reduced risk. The vessel refloated on the tide at 00:20 and was withdrawn from service. The channel was closed to commercial traffic for 19 hours.

What the investigation found

The probable cause was the continuation of the transit at about eight knots in darkness after the chart plotter froze. The vessel could have been stopped in 92 m at any point in the 114 seconds before it crossed the channel edge.

Factors that very likely contributed were:

- continuing the transit after the freeze
- the freeze itself
- the operator having no passage plan and no hold point for a night transit
- mangrove regrowth screening the No. 4 beacon, the remaining means of fixing, from the channel between 0.84 nm and 0.34 nm north of it
- the deferral of the sightline clearing at three works rounds
- the shallow alarm setting of 0.9 m under the keel, which gave 16 seconds' warning against a 92 m crash stop distance.

Four further factors likely contributed. The winter timetable moved TF18 into darkness without reopening the route risk assessment, which restricted the channel to daylight transits. The beacon's solar conversion halved its nominal range and no operator was told. The plotter's morning freezes were passed on verbally at the change of master, as the operator's rule allowed, and not entered in the defect book. An entry made that day would not by itself have stopped the service, so the rule may be a fleet weakness rather than a cause. The aids to navigation code imposed no availability, recording or notification duty on this channel. That finding is less settled, because nothing in the code prevented the Port Authority from notifying operators or clearing the sightline.

Whether new foreshore floodlighting behind the beacon hindered its detection could not be determined. The floodlights aligned with the beacon only inside the arc where the mangroves already screened it.

Low water and a steering or propulsion failure were both very unlikely. The master's duty pattern and the forward saloon lifejacket locker's position did not contribute but increased risk.

What can be done

A route assessment holds only for the operation it assessed. A timetable change that moves a service into darkness needs it reopened. A passage plan naming a primary means of fixing should say what to do when it is lost. The investigation recommended that Tarrakin Ferries write a night passage plan for the channel before any further darkness transit, requiring the way to be taken off the vessel if position fixing is lost.

An aid that cannot be seen from the water it serves is not working. On 24/07/2026 the Harbour Master trimmed the Weerona Islet canopy to 2 m and programmed a three-yearly cycle. The investigation recommended replacing the 50 m vegetation scope in the services agreement with a sightline test. A local notice giving the beacon's 3 M range was issued on 22/07/2026, and the operator reissued its passage plan with the corrected range. This action is in progress.

No safety action had been taken against four issues:

- the code's notification gap
- the timetable change process
- the verbal defect handover
- the lifejacket locker, which could not be reached from 22 of 34 forward saloon seats after a refit that triggered no re-survey.

Factual information

History of the occurrence

At about 13:05 on 07/07/2026 the masters changed over on MV Tarrakin Star at the Kanooka Heads town wharf. The outgoing master said the chart plotter had frozen twice that morning and recovered by itself both times. The relieving master replied that it had done the same to him the month before. Neither entered the fault in the wheelhouse defect book.

After two uneventful round trips, the last service, TF18, departed Muntari Island at 17:35:02 with 38 passengers, a master, a deckhand and a kiosk attendant. Sunset was at 17:02 and civil twilight ended at 17:29. The master reported the departure to Kanooka Heads Port Control on VHF channel 12, which replied that the channel was clear.

The vessel entered the Weerona Channel from the north at 18:03:58, abeam the No. 1 beacon. At 18:04:10 the deckhand asked whether to stay on the bridge, and the master sent her to collect fares. At 18:05:30 the vessel was abeam the No. 2 beacon, 1 m west of the charted centreline, at 8.4 kt on 195 degrees.

At 18:07:06 the chart plotter display froze. The chart picture stayed on screen, the vessel symbol stopped moving and the display did not respond to touch. Bridge audio captured sounds consistent with repeated screen taps at 18:07:26 and the master saying "frozen again" at 18:07:44. The plotter did not recover. The master kept the throttle settings unchanged and looked for the No. 4 starboard-hand beacon on the western side of the channel.

From 18:07:06 the course over ground increased from 195 to 200 degrees and the vessel moved west of the centreline. At 18:08:31 the master said "Right. Where's four." At 18:09:00 the vessel was 20 m west of the centreline, at the western edge of the dredged channel. At 18:09:40 the audio captured a sound consistent with a window or door being slid open. At 18:10:02 the master said "There's four. Keep her over." The vessel was then 39 m west of the centreline, outside the dredged channel and 0.25 nm north of the beacon. At 18:11:00 it was 58 m west, over Boolarong Bank.

The echo sounder shallow alarm sounded at 18:11:48 and both throttles moved to astern together at 18:11:52. At 18:12:04 the vessel contacted Boolarong Bank at 6.8 kt, 74 m west of the charted centreline and 54 m outside the dredged channel. It came to rest within 11 m, heeled 4 degrees to starboard, on a falling tide. Grinding and vibration lasted approximately 9 seconds.

The master stopped both engines at 18:12:31 and did not restart them. He addressed the passengers at 18:12:40. At 18:13:04 he broadcast a pan pan on VHF channel 16, giving the position, the persons on board, no injuries, no pollution, that he was not attempting to refloat, and requesting the workboat. Port Control acknowledged at 18:13:29 and called out the workboat. At 18:14:52 the master told the passengers to put on lifejackets.

At 18:16 the deckhand sounded the No. 2 starboard void and found approximately 75 mm of water. The master directed soundings every ten minutes and reported the ingress to Port Control at 18:17:05. At 18:22:40 the deckhand reported all passengers in lifejackets. She reported having to go around the servery, where people were standing, to reach the forward saloon locker behind the counter.

The port workboat came alongside at 18:41. The master reported all 38 passengers ashore at 19:26:08. The vessel was refloated on the rising tide at 00:20 on 08/07/2026, with the void sounding approximately 260 mm, and towed to the town wharf. The Weerona Channel was closed to commercial traffic for 19 hours.

One passenger reported a grazed shin and two were treated at the wharf for distress. There were no serious injuries.

Personnel information

Master. The master held a certificate of competency, master less than 35 m, from 2012. He had 14 years as a master, 11 on this route with Tarrakin Ferries, approximately 19 years in commercial marine operations and approximately 4,100 Weerona Channel transits. His last route check, on 14/02/2026, was satisfactory. He was on day nine of a twelve-day tour, working a split duty of 05:40 to 08:40 and 13:10 to 19:20, and had slept 6 h 15 the night before. He described himself as tired, normal for a split, and not slow.

Deckhand. The deckhand had 7 months with the operator, in her first winter season on the route, and approximately 3 years in commercial marine operations. She held a certificate of safety training and coxswain grade 2, and completed vessel familiarisation and drills in 12/2025. She worked the master's duty from 13:10 and was not on a split. She had never been shown the channel marks.

Outgoing master. The outgoing master had 9 years as a master, 4 on this route and 6 with the operator, and approximately 1,400 channel transits. His last route check, on 09/11/2025, was satisfactory. He worked 05:40 to 13:05 on the day, on day six of a twelve-day tour.

Aircraft and equipment information

MV Tarrakin Star is a 24.4 m aluminium catamaran passenger ferry certificated for 96 passengers. It completed its annual survey on 11/03/2026 with no outstanding conditions and was last slipped in 03/2025.

The defect book held eleven entries in the 90 days before the occurrence, none concerning the chart plotter, GPS antenna, echo sounder, steering gear, engines or propulsion controls. The maintenance system held no work order, fault report or correspondence on the chart plotter in 2026.

The chart plotter is a 12-inch multifunction display fed by a separate GPS antenna unit, running chart software revision 4.11 on firmware issued 03/2024. Firmware issued by the manufacturer in 11/2024 addresses cause code 0x51 and had not been applied. The operator did not hold the bulletin and had no subscription to the manufacturer's bulletins.

The echo sounder shallow alarm was set to 0.9 m under the keel, the value entered at commissioning in 2019. The trials figure for a crash stop from 8 kt is 92 m.

The vessel carries 120 adult and 20 child lifejackets in four lockers. The forward saloon locker holds 40 adult jackets in the bulkhead behind the kiosk servery counter. The counter was fitted in a 2019 refit under a minor modification notice that did not trigger a re-survey of the safety equipment arrangement. The 2011 build had an open bench there, leaving the locker face clear. The same counter is fitted to the operator's two other 24 m vessels.

Meteorological information

Both masters described the evening as clear, with a light westerly and flat water. The transit was made in full darkness.

Recorded data

The shore AIS receiver log showed the vessel within 3 m of the centreline from 18:03:58 to 18:07:06. Rate of turn peaked at 2.1 degrees per minute at 18:07:40 and returned to zero by 18:08:50, after which heading was held between 197 and 200 degrees. The set west averaged 12 m per minute.

Speed over the ground fell from 8.4 kt to 7.6 kt between 18:07 and 18:11 with no throttle movement.

The engine monitoring recorder showed both engines at 1,780 and 1,784 rpm from 18:03 to 18:11:52, with no alarm, temperature or pressure excursion, or throttle movement. Steering hydraulic pressure stayed between 96 and 99 bar against a normal band of 90 to 105 bar, and rudder angle followed helm demand within 1.5 degrees. At 18:11:52 both throttles moved to the astern detent together. Astern thrust had not reached full at contact.

The chart plotter event log recorded watchdog resets with cause code 0x51 (display process not responding) at 10:42 and 11:17 local that morning. At 18:07:06 it logged the display process not responding with no reset. Power was removed at 18:16.

The echo sounder log showed one alarm on the voyage, 16 seconds before contact, during which the vessel covered 58 m.

Aerodrome and site information

The dredged Weerona Channel has a charted half width of 20 m. The grounding position was surveyed on 08/07/2026 at 74 m west of the centreline and 34 m west of the No. 4 beacon. The bed at contact was 0.50 m below chart datum, rising to 0.15 m below datum 30 m further west, a gradient of about 1 in 70. Furrows consistent with the starboard hull and skeg ran 11 m from first contact on a heading of 197 degrees.

The No. 4 beacon was converted from mains to a solar lantern in 11/2025 under the Council's waterways energy programme. Its nominal range fell from 6 to 3 nautical miles and its character, Fl G 2.5s, was unchanged. A notification was lodged with the hydrographic service on 04/02/2026, and the chart correction was still pending on 07/07/2026. No local notice to mariners was issued and no operator was told. The operator's passage plan appendix still recorded the beacon at 6 M.

Mangrove regrowth on Weerona Islet, 240 m north-west of the beacon, was the subject of work order WO-118, raised 09/2023 and deferred as unfunded at the 2023, 2024 and 2025 works rounds. Work order WO-176, raised 03/2026, recorded the sightline as fully closed on the northern approach. The Authority's file note placed the islet outside the 50 m scope of schedule 2.4 of the services agreement. The beacon was logged as observed and operating every month from 01/2024 to 07/2026, from the town wharf 2.1 nm south, where the islet is not in the line of sight. No defect or user notice had been recorded for the beacon, and the register had never been provided to Tarrakin Ferries.

Damage

The starboard hull bottom plating was set up over 4.1 m between frames 6 and 11, with a 180 mm split in way of the No. 2 void. The starboard propeller blades were bent aft, the shaft bent 1.4 mm, the rudder stock distorted 3 degrees and the skeg torn away. The port hull, propulsion and steering were undamaged.

Tests and research

On 12/07/2026 the steering rams, tiller arms, hydraulic system and autopilot were removed and function tested. All were serviceable, and the system passed a lock-to-lock timing test.

On 21/07/2026 the chart plotter was bench tested for 46 hours with the same antenna and cabling. The fault did not recur and no defect was found.

On 18/07/2026 a check on the inbound track at the same time of night measured the No. 4 lantern at a luminous range of 2.6 nm. The light was not visible from the channel between 0.84 nm and

0.34 nm north of the beacon. The lantern is 4.2 m above chart datum. The islet canopy measured 5.8 m, against 2.1 m in a 2019 aerial survey.

On 16/07/2026 the forward saloon locker face was found unreachable from 22 of the 34 seats without passing behind the servery through a 0.6 m gap. The muster placard directs passengers to the locker forward on the port side, which is not visible from the seats.

Organisational and management information

From 01/06/2026 the winter timetable placed the evening returns in darkness, and TF18 was the thirty-first such transit. Both masters and the deckhand stated that the change brought new times or a new roster sheet and nothing else. There was no night passage plan for the channel. The passage plan named a speed, a course and three beacons and said nothing about bridge manning. Procedure OPS-3 is silent on a second person on the bridge.

Both masters stated that a fault that had cleared was passed on verbally and not entered in the defect book, and the master stated that this was common practice. The outgoing master believed the plotter had frozen three or four times in the past year without being reported. Both masters had noticed the No. 4 beacon becoming harder to see over two years and had never reported it. The company reporting system has a "navigation hazard" category, unused on this route in three years, and no master interviewed knew it existed. No operator had asked the Port Authority about the channel's aids.

Analysis

External Influences

Aids to navigation on this waterway carry no availability standard and no duty to tell users anything

Section 4.2 of the Waterways Aids to Navigation Code WAN-2 places three duties on the authority responsible for an aid on a declared shipping channel. It must record the aid's availability. It must achieve an availability of "not less than 99.0 per cent measured annually". It must notify a change to the aid's "character, position or range by notice to mariners before the change takes effect".

The Weerona Channel is not a declared shipping channel, so section 4.3 applied instead. Under section 4.3 the Authority needed only to show that each aid was inspected "at intervals it has determined and recorded".

The services agreement between Kanooka Shire Council and the Authority set those inspection intervals:

- a daylight inspection once a quarter
- an observation from the town wharf once a month.

The agreement also limited vegetation work to within 50 m of an aid. It stated that there was "no requirement in this agreement to notify waterway users of a defect, a deferral or a change to an aid."

The Authority complied with both the code and the agreement. It was virtually certain that the framework existed in this form. It is set out in the documents themselves, and the channel's status was not in dispute.

Under this framework, nothing carried the No. 4 beacon's condition to the people using the channel. The code contains three mechanisms that would have done so: a recorded availability figure, a measured range and a notice to mariners. Each of them stopped at the channel's boundary. In their place, information about the beacon passed only informally between masters [NOT ESTABLISHED: the occasion, participants and content of any exchange between masters about the No. 4 beacon].

The solar conversion in 11/2025 halved the beacon's nominal range from 6 to 3 nautical miles.

- A notification was lodged with the hydrographic service on 04/02/2026, but the chart correction was still pending on 07/07/2026.
- No local notice to mariners was issued.
- The operator's passage plan appendix still recorded the beacon at 6 M on the night.

Had section 4.2 applied, the range change would have been notified to users before it took effect. That duty was the only mechanism identified that would have put the change in front of the operator.

The agreement also shaped how the beacon was checked and maintained:

- **Monthly check.** The beacon was recorded as "observed, operating" every month from 01/2024 to 07/2026. Each check was made from the town wharf, 2.1 nm south of the beacon, where Weerona Islet is not in the line of sight. The checks complied with the agreement but could not reveal screening on the northern approach.
- **Vegetation limit.** The 50 m limit placed the islet, 240 m from the beacon, outside the aids to navigation scope. Its clearing therefore competed in the general waterways vote with boat ramps and pontoons.

The Organisational Influences subsection argues how the conversion and the deferred clearing came about.

There was evidence against this factor's influence:

- Nothing in the code stopped the Authority from notifying operators, trimming the mangroves or measuring the light. It did not do so for reasons of money and priority, not because the code allowed it.
- A 99.0 per cent availability target would not have detected the problem. The lantern was lit and working throughout, and the problem was that it could not be seen from the channel. No availability figure measures that.
- The notification duty in section 4.2 covers character, position and range. It does not, on its terms, cover an obstructed sightline.

On this reading, the code's scope was background to the occurrence rather than a cause of it.

The investigation gave that argument weight and accepted the factor's influence on narrower grounds only:

- The availability target and the recording duty would not have revealed the screening.
- The notification duty would very probably have revealed the range change, which the operator's own passage plan showed it did not know about.

On that basis, the influence of the code's threshold was assessed as likely. That is a lower level of confidence than the other contributing factors in this investigation, and the matter is not presented as settled.

No party was in breach of any requirement. The issue lies in the threshold itself. It leaves a 40 m wide channel used by a 96-passenger ferry with beacons that no one is required to measure, record or report on.

It is likely that the absence of any availability, recording or notification duty for aids on the Weerona Channel contributed to the operator transiting the channel without knowing of the change to the No. 4 beacon's range.

Organisational Influences

Winter timetable moved the last service into darkness without reopening the route risk assessment

RRA-07 issue 2 was the current route risk assessment for the Weerona Channel. It stated: "Operating restriction: daylight transits only. A night transit of the Weerona Channel requires the General Manager Operations' approval and a specific passage plan." CHG-2026-014 moved TF18 from 16:15 to 17:35 from 01/06/2026. It assessed the change as "timetable change only", recorded "No new hazard identified" and answered "No" to "Route risk assessment review required". The Operations Manager approved it. That is one level below the approver the restriction names. RRA-07 was not reopened and no night passage plan was written. The General Manager Operations' approval was neither sought nor given. Both documents are current controlled copies, and it was virtually certain that the change was made in this way.

The change process asked whether anything about the service was new. The route, vessel, crew qualifications and berths were not new. The process did not ask whether the existing assessment still held for the hours in which the service would now run. The assessment already named a "single unlit reach between the No. 2 and No. 4 beacons" as a hazard. A review against a night operation would have reached that reach. Under the assessment's own terms, it would have required a night passage plan before the first service ran. Instead, TF18 made 31 transits in darkness under a plan

whose only reference to darkness was the statement that night transits were not scheduled. The master had made 26 of those transits.

Two matters limit the strength of this conclusion. A review might have produced a night plan that said no more than the daylight plan. The Operations Manager's understanding of what the restriction required was given only verbally and was not tested in formal interview. That uncertainty bears on why the change was approved as it was, not on its effect.

Approving the winter timetable without reviewing RRA-07 likely led to the absence of a night passage plan, and of any hold point for loss of the primary means of fixing, which is examined under Risk Controls.

Beacon converted to solar, nominal range halved, and no operator was told

The No. 4 beacon was converted to a solar lantern in 11/2025. The new lantern is rated at a nominal 3 nautical miles, against 6 for the lantern it replaced. The character, Fl G 2.5s, did not change. A notification reached the hydrographic service on 04/02/2026, and the chart correction was still pending on the day of the occurrence. No local notice was issued and no operator was told. The aids to navigation register had never been provided to Tarrakin Ferries. Appendix A issue 2 still gave "No. 4 (Fl G 2.5s, range 6 M)". The investigator measured a luminous range of 2.6 nm on 18/07/2026. All of this is documentary or independently measured, and it was virtually certain.

The plan named the No. 4 beacon as part of the secondary means of fixing. It described a light with more than twice the nominal range of the lantern actually fitted. The master began looking for the beacon at 18:08:31, 0.47 nm north of it, and called it at 18:10:02, 0.25 nm north of it. That was 91 seconds later. A weaker light than expected is harder to acquire.

Most of that search took place inside the arc from 0.84 nm to 0.34 nm north of the beacon, where the light was screened entirely. Over that stretch, no lantern rating would have made it visible. This is why the influence of the reduced range was assessed below that of the screening. The effect of background lighting on detection was examined separately and could not be resolved. Nothing from that matter is carried here.

The halving of the No. 4 beacon's nominal range, without notice to the operator, likely contributed to the beacon not being seen from the channel over the half mile before the vessel left it.

Clearing of the beacon's sightline deferred unfunded for three works rounds

Work order WO-118 was raised in 09/2023. It recorded mangrove regrowth on Weerona Islet screening the No. 4 beacon from the northern approach, over about half a mile to a quarter mile. It recommended trimming to 2 m on a three-yearly cycle, at an estimated \$18,500. The work was deferred as unfunded at the 2023, 2024 and 2025 works rounds. WO-176, raised in 03/2026, recorded the sightline as fully closed on the northern approach. The Authority's file note explained the funding position. The islet is 240 m from the beacon, outside the 50 m scope of schedule 2.4 of the services agreement. The work therefore competed with boat ramps and pontoons in the general waterways vote, not with aids to navigation. The 2019 aerial survey gave the canopy at 2.1 m. It measured 5.8 m on 18/07/2026, against a lantern height of 4.2 m. It was virtually certain that the deferral occurred.

The Authority identified the hazard correctly almost three years before the occurrence. It recorded the sightline's full closure about four months before. The funding scope, measured by distance from the structure and not by the aid's purpose, kept the work out of the aids to navigation vote. The monthly check did not reveal the problem. It was made from the town wharf, 2.1 nm south, where the islet is not in the line of sight. From there the beacon was recorded as observed and operating every month from 01/2024 to 07/2026, and no defect was ever recorded against it.

The vessel crossed the channel's western edge at about 18:09, 0.40 nm north of the beacon. That is inside the screened arc. The light was not dim at that point. It was absent from the master's view during the interval in which stopping or correcting within the channel was still possible. The remaining doubt is that the master might not have acquired the light in that minute and a half even had it been visible.

Deferring the clearing of the No. 4 beacon's sightline over three works rounds very likely contributed to the beacon not being visible from the channel over the half mile before the vessel left it.

Risk Controls

No night passage plan for the channel, and no hold point if the primary means of fixing is lost

The procedure under which TF18 transited the channel was written for daylight. OPS-3 3.4 requires the transit to be "made on the charted centreline, monitored on the chart plotter, with the No. 2 and No. 4 beacons identified visually before the vessel passes the No. 1 beacon." Appendix A names the plotter as the primary means of position fixing. It names the visual transit of the No. 4 beacon and the town wharf leading lights as the secondary means. Its only reference to darkness is its closing line: "Night transits are not scheduled on this route."

The plan has no night version. It names no abort criterion and no hold point, and it gives no guidance on what to do if the plotter is lost. OPS-3 3.6 requires a reduction in speed and consideration of anchoring, but only "in reduced visibility", which did not apply. The secondary means was also weaker than it appeared. The leading lights lie 2.1 nm beyond the No. 6 beacon and do not resolve as two lights until the vessel is inside the channel's southern third. When the plotter froze, the No. 4 beacon was the whole of the secondary means. The absence of a night plan followed directly from the winter timetable change, which is discussed under Organisational Influences. It was virtually certain that this gap existed.

The master had 114 seconds between the freeze at 18:07:06 and the vessel crossing the channel's western edge at about 18:09. The water was flat and there was no other traffic. A named hold point would have been available throughout that interval. Both masters interviewed told the investigation, independently, that stopping was the right response and that nothing in the operator's procedures required it.

One consideration weighs against the link. A master who did not think of stopping without prompting might not have applied a written rule under the same pressure. That possibility was weighed and did not displace the conclusion. It is why the link is put at very likely rather than higher. How the master conducted the transit after the freeze is analysed under Individual Actions.

The absence of a night passage plan, and of any hold point on loss of the plotter, very likely contributed to the transit continuing at 8 knots after the plotter froze.

Change-of-master defect brief is verbal, with nothing checking that what is said gets written down

OPS-9 requires immediate reporting only for a defect that "renders the vessel unseaworthy or which affects a statutory item of equipment." The chart plotter is not a statutory item on a class 1D vessel on this route, although it is the primary means of position fixing in the passage plan. All other defects fall under 9.2. They are entered "at the end of the shift in which they are found" and reviewed weekly. For a fault that clears itself, that is no trigger at all. Under 9.3 the outgoing master

briefs the relieving master on "any defect". Nothing requires the brief to be written down, and nothing requires the relieving master to confirm an entry was made.

Both masters independently described the same practice. The defect book was for what was broken at handover, and faults that had cleared were spoken about. The outgoing master believed the plotter had frozen three or four times on this vessel in the past year, to him and to other masters, and that none of those freezes had been written up. The defect book held eleven entries in the 90 days before the occurrence and none concerned the plotter. The maintenance system held no plotter entry of any kind for 2026. It was virtually certain that the practice existed. It explains why the morning's two freezes went unrecorded. It does not excuse the omission.

The investigation weighed two readings of the influence.

- On the first, the rule kept a recurring fault on the primary means of fixing invisible to everyone ashore. Over a year, that meant the firmware issued in 11/2024 to address cause code 0x51 was never applied. It also meant the master met the fault on the night with no reason to treat it as more than a nuisance.
- On the second, an entry made at 13:05 would have been reviewed at the end of that week, not that afternoon, and TF18 would have sailed either way. A different outcome would have required a same-day decision that nobody was required to make. On this reading the rule is a fleet finding and not a cause of this grounding.

The first reading was preferred. It rests on the accumulated history of unreported freezes rather than on the single entry not made that day.

It was likely that the verbal change-of-master brief, with no check that defects were written down, contributed to the plotter fault being passed on verbally and not entered in the defect book. The mechanism runs through the cumulative failure of recurring freezes to reach anyone who could act, not through the omission on the day alone.

Forward saloon lifejacket locker not reachable from most of the saloon it serves

The forward saloon's locker holds 40 adult lifejackets and is mounted behind the kiosk server counter. The investigation measured the arrangement on 16/07/2026. The locker face cannot be reached from 22 of the 34 forward saloon seats without passing behind the counter through a 0.6 m gap. With three people standing at the counter, as there were on the night, that gap is the only route. The muster placard directs passengers to a locker they cannot see from their seats. The deckhand reported at the time: "There's two in the forward saloon I had to go round the server for, the locker's behind the counter and there's people standing at it." It was virtually certain that the condition existed.

The locker played no part in the grounding. All 38 passengers were in lifejackets by about 18:22 and ashore by 19:26. It was unlikely that the locker influenced the outcome.

It was very likely nonetheless that the arrangement increased risk. The vessel was aground, upright and stable, so the muster could be unhurried. In a fire, a night evacuation into the water, or with a deckhand injured in the impact, passengers who cannot reach their lifejackets without crew help would have to wait for them. The underlying gap lies in the modification route. The 2011 approved arrangement had an open bench with the locker face clear. The counter was fitted in a 2019 refit under a minor modification notice, which did not trigger a re-survey of the safety equipment arrangement. The same counter is fitted to the operator's two other 24 m vessels.

A modification route that does not review safety equipment allowed the forward saloon locker to become unreachable from most of the saloon it serves, on this vessel and very likely across the operator's 24 m vessels. (Safety issue)

Local Conditions

Chart plotter display froze during the channel transit and did not recover

Three independent sources place the freeze at 18:07:06.

- **Event log.** The plotter's event log recorded the display process not responding, with no reset.
- **Bridge audio.** The audio matches to the second: "Oh, come on" at 18:07:11, sounds consistent with repeated screen taps at 18:07:26, and "(unintelligible) frozen again" at 18:07:44.
- **Master's account.** The master described the chart picture held on screen, the vessel symbol stationary, and the display not accepting touch input.

The same cause code, 0x51, had been logged at 10:42 and 11:17 that morning. On both of those occasions the unit recovered itself.

The 46 hour bench test on 21/07/2026 did not reproduce the fault and found no defect. That result goes to why the display froze, not to whether it did. The contractor identified cause code 0x51 as known to the manufacturer and addressed by firmware issued in 11/2024. That firmware had not been applied to this unit.

The freeze removed the primary means of position fixing named in the passage plan. The vessel held the centreline while the plotter worked and left it within two minutes of the freeze. The master's search for the beacon began 85 seconds after it. The decision to continue the transit after the freeze is considered separately.

It was virtually certain that the plotter display froze at 18:07:06 and did not recover, and extremely likely that this contributed by removing the primary means of fixing for the rest of the transit.

No. 4 beacon not visible from the channel over the half mile before the vessel left it

The range and visibility check on 18/07/2026 was made from the port workboat on the inbound track at the same time of night. It found that the No. 4 beacon cannot be seen from the channel between 0.84 nm and 0.34 nm north of it. Over that stretch the sightline passes through the mangrove canopy on Weerona Islet, 240 m north-west of the beacon. The canopy was measured at 5.8 m. The lantern is 4.2 m above chart datum.

The master's search fell entirely inside that screened arc:

- The vessel was 0.68 nm north of the beacon when the plotter froze.
- It was 0.47 nm north at 18:08:31, when the master said "Right. Where's four".
- It was 0.25 nm north at 18:10:02, when he said "There's four. Keep her over".

A window or door was heard being slid open at 18:09:40. The vessel crossed the channel's western edge at about 18:09, between the two calls. Both masters raised the beacon unprompted in interview. Neither had ever reported it.

The measurement was made eleven days after the occurrence, and the canopy was not surveyed on the night. That is the only reservation about whether the screening existed.

The vessel had begun to set west before the master started looking, so the screening does not account for the start of the departure. With the plotter frozen, however, the beacon was the only means of fixing available, and it was absent for the whole period in which the vessel set west. The light was not missed through any lapse of lookout. Over that stretch there was nothing to see. The

solar conversion and the deferred clearing of the sightline, which together produced this condition, are analysed under Organisational Influences.

The No. 4 beacon was extremely likely not visible from the channel over the stretch in which the master searched for it, and this very likely contributed to the vessel leaving the dredged channel without the departure being detected.

New foreshore floodlighting behind the beacon reduced the master's ability to pick it up

The investigation examined whether background lighting, rather than the mangroves alone, prevented the master from seeing the light.

The evidence for the hypothesis was as follows:

- **Independent accounts.** Both masters described the beacon as lost against shore lighting. The master referred to the town and the new car park. The outgoing master described the light as washed out from the northern approach.
- **The floodlights.** The lighting survey of 18/07/2026 found 14 unshielded LED floodlights on 12 m columns at the new boat ramp. Each presents about 4,800 candela toward the channel. They were commissioned on 22/05/2026, ten days before the winter timetable began.
- **The alignment.** From 0.5 nm north of the beacon, the array lies behind it within 4 degrees of the same bearing and within 1.5 degrees of the same elevation. The array is continuously lit, while the beacon flashes green for 0.4 seconds every 2.5 seconds.
- **No assessment.** The Port Authority's aids to navigation officer was not consulted on the development. No assessment against the channel's aids was required or made.

The evidence against the hypothesis was as follows:

- **Position of the alignment.** The one position at which the alignment was measured lies inside the screened arc. At that position the beacon would not have been visible against a dark background either.
- **The sighting.** The master sighted the light inside the arc's southern limit, with the floodlights in the same place and no dimmer.
- **No measurement on the night.** No measurement exists of what could be seen on the night.
- **No detection threshold.** Nothing establishes a detection threshold for a 0.4 second green flash against that background.

The evidence shows a real alignment. It does not show that the alignment affected detection. While the mangroves screen the beacon over the same stretch, the two effects cannot be separated. Separating them would require the light's visibility against the floodlights to be measured with the sightline clear.

Whether the floodlighting reduced the master's ability to detect the beacon was assessed as about as likely as not. The matter is unresolved, and nothing from it is carried into the findings.

Water level below prediction reduced the depth over Boolarong Bank

This hypothesis was very unlikely. At 18:12 the Kanooka Heads gauge read 0.91 m against a predicted 0.94 m. That left 1.41 m of water over the bed at the contact point, against a static draught of 1.35 m and a calculated squat of 0.18 m. The vessel grounded because it was 54 m outside a channel dredged to 3.2 m, not because the water was lower than predicted.

Day nine of twelve, on a split duty across a thirteen hour forty spread

It was virtually certain that the master was on his ninth consecutive day. His split duty ran from 05:40 to 08:40 and from 13:10 to 19:20, a spread of 13 hours 40 minutes. He had recorded 6 hours 15 minutes of sleep the night before.

Influence on this occurrence was unlikely, because the recorded data contradicted each mechanism through which fatigue would have had to act:

- The helm held the centreline to within 3 m while the plotter worked.
- The response to the shallow alarm took 4 seconds, against a fleet figure of 6 to 9 seconds.
- The radio work after the grounding was complete and correct at the first attempt.
- The decision not to refloat was made in about thirty seconds and was correct.
- On the one decision not made, to stop, both masters independently said that nothing in any document prompts that consideration.

The exposure is nonetheless likely to matter. Every darkness transit of the winter timetable falls at the end of such a duty, late in the cycle. No relief master is rostered, and the fatigue diary the operator asks masters to keep is never collected.

Individual Actions

Plotter fault passed on verbally at the change of master and not entered in the defect book

The plotter's event log recorded two freezes during the morning shift, at 10:42 and 11:17. The unit recovered from each one by itself.

At the change of master at about 13:05, the outgoing master told the relieving master about both freezes. The relieving master said the unit had done the same to him the month before. Neither master entered the fault in the defect book or reported it to the Operations Manager. Their accounts agree in every particular, and both volunteered the omission before being asked. It was virtually certain that the fault was passed on verbally and not entered.

The omission was consistent with the procedure as written. OPS-9 at 9.3 requires the outgoing master to brief the relieving master on "any defect". It does not require the brief to be written, and it does not require the relieving master to confirm that a defect was entered. The plotter is not a statutory item, so 9.2 applied. That item requires entry "at the end of the shift in which they are found", with review by the Operations Manager "weekly". For an intermittent fault that clears itself, this provides no trigger. The weakness lay in the change-of-master practice, which is examined under Risk Controls, and not in the two masters.

The effect of the missing entry extended beyond one evening. The maintenance system held no plotter entry for 2026. The operator held no subscription to the manufacturer's bulletins, and the firmware issued in 11/2024 that addresses cause code 0x51 had not been applied. An entry would have put a recurring fault in front of the people who obtain firmware and take vessels out of service. Against this, a single entry made at about 13:05 would have been reviewed on a weekly cycle. It would not by itself have stopped TF18 sailing that evening.

The absence of a defect book entry for the morning's plotter freezes likely contributed to the uncorrected fault remaining on the vessel for the evening transit.

Transit continued at eight knots after the plotter froze, rather than stopping in the channel

The plotter froze at 18:07:06, and the AIS track shows speed held at 8.4 kt to 18:08. The later fall to 7.6 kt occurred with no throttle movement and is accounted for by shoaling drag. At 18:08:31, 85 seconds after the freeze, the bridge audio records "Right. Where's four". It records nothing about stopping. The master's own account agrees. He did not consider stopping and turned directly to the visual transit, which he regarded as the backup. It was virtually certain that the transit continued without reduction of speed.

Nothing in the conditions prevented a stop. There was no other traffic, the water was flat and the wind was offshore at 8 kt. The vessel could be stopped in 92 m. It crossed the channel's western edge 114 seconds after the freeze, and a stop at any point in that interval would have left it in the dredged channel. Continuing extremely likely contributed to the vessel leaving the channel and to the shoaling going undetected until the shallow alarm.

The master made the decision without procedural support. Appendix A names the visual transit of the No. 4 beacon as the secondary means of fixing, so his turn to the beacon followed the plan. The plan names no hold point and gives no guidance on losing the plotter. OPS-3 at 3.6 addresses reduced visibility, not the loss of the primary means of fixing. Those gaps are examined under Risk Controls.

The beacon was not visible from the channel over that stretch, as set out under Local Conditions. No lookout was on the bridge. The deckhand had been sent to collect fares at 18:04:14, which was normal practice on the route and is not addressed in OPS-3.

Continuing at 8.4 kt after the plotter froze extremely likely contributed to the grounding, and the decision was made under a passage plan with no hold point while the beacon sought was screened from view.

Engines stopped and no attempt made to refloat on a falling tide

Both engines were stopped at 18:12:31, 27 seconds after contact, and were not restarted. In the pan pan at 18:13:04 the master stated "I am not attempting to refloat" as a decision already made. He then had the No. 2 starboard void sounded and reported about 75 mm of ingress at 18:17:05. He kept the void under soundings every ten minutes after that.

The master addressed the passengers 36 seconds after the impact and again at 18:14:52. Lifejackets were issued to all 38 passengers. In interview, he gave his reasons as the falling tide, the unknown bottom, the number of people aboard, and that the vessel could not sink while it was on the bank. It was virtually certain that these actions occurred as described.

The survey confirmed a 180 mm split in the starboard plating in way of the No. 2 void. Ingress reached about 260 mm by the time the vessel refloated at 00:20. Backing off into the channel, with a minimum depth of 3.2 m below datum and the ingress unassessed, would have put a flooding compartment into deep water at night with the passengers still aboard. The actions were taken quickly and without hesitation.

Stopping the engines and not attempting to refloat a holed vessel on a falling tide very likely reduced the risk to the 38 passengers and three crew.

Occurrence Events

Vessel left the dredged channel and tracked onto Boolarong Bank

The AIS track placed the vessel 1 m west of the charted centreline abeam the No. 2 beacon at 18:05:30. It held the centreline until 18:07:06. From then the heading increased steadily from 195 to 200 degrees over about a minute and was held. The vessel moved west at an average of 12 m per minute. It reached the western edge of the 40 m channel at about 18:09, was 39 m west of the

centreline at 18:10:02 and was over Boolarong Bank from about 18:11. The surveyed grounding position, 74 m west of the centreline and 54 m outside the channel, agreed with the track. So did the furrows in the bed. The departure from the channel was virtually certain.

The westward movement was not a current or wind effect. The recorded rate of turn peaked at 2.1 degrees per minute at 18:07:40 and was back to zero by 18:08:50. That shape is a heading change applied and then held. The wind was offshore at 8 kt, and there was no cross-set in the channel at that state of tide. The heading change began within a minute of the chart plotter freeze at 18:07:06.

The channel is dredged to 3.2 m. Every metre of the vessel's 74 m of westing took it further from that depth. The grounding followed directly, and only, from the vessel being outside the dredged channel.

The reasons the new heading was steered and held are argued in other subsections. The screening of the No. 4 beacon is argued under Local Conditions. The continuation of the transit after the plotter froze is argued under Individual Actions. The floodlighting behind the beacon was examined but could not be resolved, and it is discussed in its own subsection.

It was virtually certain that the vessel left the dredged channel on a steered and held heading, and virtually certain that this resulted in the grounding.

Shoaling not detected until sixteen seconds before contact

The echo sounder's shallow alarm was set to 0.9 m under the keel. The alarm log records this as "the value written into the setup when the unit was commissioned in 2019 and it has not been changed." The alarm sounded once, at 18:11:48. The bridge audio records the tone beginning at that second. Both throttles were moved to astern together 4 seconds later. Contact followed at 18:12:04. The alarm log, the bridge audio and the recorded speed agree to the second. The late detection was virtually certain.

The equipment did what it was set to do. The shortfall lay in the setting and the speed. At 7.1 kt the vessel covered 58 m in the 16 seconds. The vessel's own trials figure for a crash stop from 8 kt is 92 m. On the surveyed bank gradient of about 1 in 70, the alarm "could not have sounded appreciably earlier at that setting." Astern thrust was developing but had not reached full at the moment of contact, and the vessel struck at 6.8 kt.

The warning margin was therefore fixed before the alarm sounded. A 0.9 m setting on a 1 in 70 gradient at eight knots gave less warning than the vessel's stopping distance. A setting that gave about forty seconds would have changed the outcome. So would a transit speed at which sixteen seconds was enough. With either, the vessel would not have touched. The transit speed itself is argued under Individual Actions.

It was virtually certain that shoaling was not detected until 16 seconds before contact, and extremely likely that this contributed to the grounding, because the alarm setting at the transit speed gave less warning than the vessel needed to stop.

Steering or propulsion failure caused the vessel to sheer off track

A steering or propulsion failure was very unlikely to have existed and was ruled out on measurement. The engine recorder showed both engines at 1,780 and 1,784 rpm with no alarm, excursion or throttle movement until 18:11:52. Steering hydraulic pressure held at 96 to 99 bar against a normal band of 90 to 105 bar. Rudder angle followed helm demand within 1.5 degrees. The rate of turn showed a heading applied and held rather than a sheer. When the steering system was removed and function tested on 12/07/2026, every component was serviceable, with no sign of a pre-existing defect.

Conclusions

These findings are made with respect to the occurrence and should not be read as apportioning blame or liability to any organisation or individual.

A finding stated without a likelihood term was established as fact.

Findings

1. The Waterways Aids to Navigation Code WAN-2 applied its availability, recording and notification duties only to aids on a declared shipping channel, and because the Weerona Channel was not one, nothing required the No. 4 beacon's range or visibility to be measured, recorded or notified to users. This likely contributed by leaving no means of putting the beacon's condition before the operator, although nothing in the code prevented the Kanooka Heads Port Authority from notifying operators or clearing the sightline, and an availability target would not have detected a light that was lit but screened. (Safety issue)
2. Change record CHG-2026-014 moved service TF18 into darkness from 01/06/2026 as a timetable change only. Route risk assessment RRA-07 was not reopened, although it restricted the channel to daylight transits and required the General Manager Operations' approval and a specific passage plan for any night transit. This likely contributed by removing the review that would have reached the unlit reach between the No. 2 and No. 4 beacons. (Safety issue)
3. The No. 4 beacon was converted to a solar lantern in 11/2025, its nominal range fell from 6 to 3 nautical miles, and no local notice to mariners was issued and no operator was told, so the operator's passage plan still described a 6 M light. This likely contributed, although over the stretch where the master searched for it the light was screened entirely, and a brighter light would not have been visible there either. (Safety issue)
4. Clearing of the mangrove regrowth on Weerona Islet that screened the No. 4 beacon from the northern approach was deferred as unfunded at the 2023, 2024 and 2025 works rounds, because the islet lay outside the 50 m scope of the services agreement. This very likely contributed by leaving the beacon's sightline closed when the vessel needed it. (Safety issue)
5. The operator had no night passage plan for the Weerona Channel, and its passage plan named the chart plotter as the primary means of position fixing but gave no hold point, abort criterion or guidance for losing it. This very likely contributed by leaving the master without a named action to stop in the channel when the plotter froze. (Safety issue)
6. The operator's change-of-master defect brief was verbal, and nothing required its content to be written down or the relieving master to check that a defect mentioned in it had been booked. This likely contributed by keeping an intermittent fault on the primary means of position fixing from anyone ashore, so that the manufacturer's 11/2024 firmware addressing it was never applied. A defect book entry made on the day would not by itself have stopped TF18 sailing, and on that reading the practice is a fleet finding rather than a cause of this grounding. (Safety issue)
7. The forward saloon lifejacket locker could not be reached from 22 of the 34 forward saloon seats without passing behind the kiosk counter. The counter was fitted in a 2019 refit under a minor modification notice that did not trigger a re-survey of the safety equipment arrangement, and it is also fitted to the operator's two other 24 m vessels. The arrangement was unlikely to have contributed to the occurrence, and it very likely increased risk by degrading passengers' ability to equip themselves in an emergency. (Safety issue)

8. The chart plotter display froze at 18:07:06 during the channel transit and did not recover, with the same cause code logged at 10:42 and 11:17 that morning. This very likely contributed by removing the primary means of position fixing named in the passage plan.

9. It was very likely that the No. 4 beacon was not visible from the channel between 0.84 nm and 0.34 nm north of it, because the mangrove canopy on Weerona Islet stood at 5.8 m against a lantern height of 4.2 m. This very likely contributed because the beacon was the only remaining means of fixing while the vessel set west.

10. It was very unlikely that the water level was below prediction to a degree that reduced the depth over Boolarong Bank. The port gauge read 0.91 m at 18:12 against a predicted 0.94 m, and the vessel touched because it was 54 m outside the dredged channel.

11. The master was on day nine of a twelve-day tour, on a split duty of 9 hours 10 minutes across a spread of 13 hours 40 minutes, within the operator's limits and the state crewing determination. The recorded evidence showed no sign of fatigue on the night, and this was unlikely to have contributed to the occurrence. The roster likely increased risk on every winter darkness transit of the route.

12. The two chart plotter freezes on the morning of 07/07/2026 were passed on verbally at the change of master at about 13:05 and were not entered in the defect book. This likely contributed by keeping the recurring fault from the people who could have applied the firmware fix or withdrawn the vessel, although a single entry reviewed on a weekly cycle would not by itself have stopped TF18 that evening.

13. The transit continued at about eight knots after the chart plotter froze, with no consideration of stopping recorded, although the vessel could have been stopped in 92 m on flat water with no other traffic at any point in the 114 seconds before it crossed the channel's western edge. This very likely contributed to the vessel leaving the channel.

14. The master stopped both engines 27 seconds after contact and made no attempt to refloat a holed vessel on a falling tide with 38 passengers aboard. This very likely reduced risk by keeping a flooding compartment out of deep water.

15. From 18:07:06 the vessel's heading was increased from 195 to 200 degrees and held, so that the vessel left the dredged channel at about 18:09:00 and tracked onto Boolarong Bank, coming to rest 74 m west of the charted centreline. This resulted in the grounding.

16. The echo sounder shallow alarm, set at 0.9 m under the keel since commissioning in 2019, sounded at 18:11:48, 16 seconds before contact, against a crash stop distance from 8 kt of 92 m. This very likely contributed by giving less warning than the vessel needed to stop at its transit speed.

17. It was very unlikely that a steering or propulsion failure caused the vessel to leave the channel. The engine and steering recorder traces were normal throughout, the heading change was applied and held rather than a sheer, and every steering component was found serviceable on examination.

Probable cause

The probable cause of the grounding was the continuation of the channel transit at about eight knots in darkness after the chart plotter display froze, on a night operation for which the operator had no passage plan and no hold point, while the No. 4 beacon, the remaining means of fixing, was screened from the channel by mangrove regrowth whose clearing had been deferred. This resulted in the vessel leaving the dredged channel and tracking onto Boolarong Bank, with too little warning from the shallow alarm to stop.

Contributing factors

- The winter timetable moved the last service into darkness without reopening the route risk assessment, and this likely contributed. (Safety issue)
- No night passage plan existed for the channel, and no hold point was set for loss of the primary means of fixing, which very likely contributed. (Safety issue)
- The change-of-master defect brief was verbal, with nothing checking that what was said got written down. It likely contributed, although a written entry on the day would not by itself have stopped the service. (Safety issue)
- The plotter fault was passed on verbally at the change of master and not entered in the defect book, which likely contributed.
- The chart plotter display froze during the channel transit and did not recover. This very likely contributed.
- After the plotter froze the transit continued at eight knots rather than stopping in the channel, and this very likely contributed.
- Aids to navigation on this waterway carried no availability standard and no duty to tell users anything. This likely contributed, although the code did not prevent notification or clearing. (Safety issue)
- The No. 4 beacon was converted to solar, its nominal range halved, and no operator was told. This likely contributed. (Safety issue)
- Clearing of the beacon's sightline was deferred unfunded for three works rounds, which very likely contributed. (Safety issue)
- It was very likely that the No. 4 beacon was not visible from the channel over the half mile before the vessel left it, and this very likely contributed.
- The vessel left the dredged channel and tracked onto Boolarong Bank, which resulted in the grounding.
- Shoaling was not detected until sixteen seconds before contact, and this very likely contributed.

Other factors that increased risk

- The forward saloon lifejacket locker was not reachable from most of the saloon it served. It was unlikely to have contributed to this occurrence, and it very likely increased risk to passengers in an emergency. (Safety issue)
- The master was on day nine of twelve, on a split duty across a thirteen hour forty spread. This was unlikely to have contributed to this occurrence, and the roster likely increased risk on the route's winter darkness transits.

Other findings

- Water level below prediction reducing the depth over Boolarong Bank was examined and ruled out. It was very unlikely to have occurred, because the gauge stood 0.03 m below the predicted height at the time of the grounding.
- A steering or propulsion failure causing the vessel to sheer off track was examined and ruled out. Such a failure was very unlikely, given the recorder traces and the serviceable steering components.
- The master stopped the engines and made no attempt to refloat on a falling tide, and this very likely reduced risk.

- Whether new foreshore floodlighting behind the beacon reduced the master's ability to pick it up could not be determined either way. Fourteen unshielded floodlights, commissioned on 22/05/2026, lay within 4 degrees of the beacon's bearing from 0.5 nm north, and both masters independently described the light as washed out against them. Against this, that position lies inside the arc where the mangroves screened the beacon entirely, and the master picked up the light at 0.34 nm with the floodlights unchanged. No measurement exists of what could be seen on the night, and no detection threshold has been established for the beacon's flash against that background. The matter could not be settled while the mangroves screen the light over that arc.

Recommendations

Aids to navigation on this waterway carry no availability standard and no duty to tell users anything

The Waterways Aids to Navigation Code WAN-2 applies its availability target, its duty to record availability and its duty to notify a change of character, position or range by notice to mariners only to aids on a declared shipping channel. The Weerona Channel is not a declared shipping channel. Its beacons fall under section 4.3, which requires only that the responsible authority can demonstrate an inspection interval it has set itself. The services agreement sets that interval at a quarterly daylight inspection and a monthly observation from the town wharf. It limits vegetation work to 50 m from the aid. It imposes no duty to tell any waterway user of a defect, a deferral or a change. The Port Authority was not in breach of either instrument.

It was likely that this gap contributed to the grounding. The notification duty in section 4.2 was the one mechanism that would have put the halving of the No. 4 beacon's nominal range in front of the operator. The operator's passage plan still carried the old range on the night. This finding is less certain than the report's other findings on contribution. The code did not prevent the Port Authority from notifying operators, clearing the sightline or measuring the light. An availability target would also not have detected a beacon that was lit but could not be seen. The code's threshold nonetheless leaves a 40 m wide channel used by a 96-passenger ferry with beacons that no one is required to measure, record or report on.

No safety action has been recorded against this safety issue.

Beacon converted to solar, nominal range halved, and no operator was told

The No. 4 beacon was converted from a mains supply to a solar lantern in 11/2025. The conversion halved its nominal range from 6 to 3 nautical miles. No local notice to mariners was issued and no operator using the channel was told. A notification was lodged with the hydrographic service on 04/02/2026, and the chart correction was still pending on 07/07/2026.

The issue mattered because the master was navigating to a written passage plan that described a light more than twice as strong as the one in the beacon. That light also stood against town and foreshore lighting added in the same period. The weaker light likely contributed to the No. 4 beacon not being visible from the channel over the half mile before the vessel left it.

On 22/07/2026, local notice to waterway users LNM-2026-14 was issued. It gave the beacon's solar conversion and its 3 M nominal range. The Port Authority's aids to navigation register was made available to every commercial operator using the channel. The action was led by Kanooka Shire Council. Tarrakin Ferries has reissued the Weerona Channel passage plan appendix as issue 3 with the corrected range. This action was recorded as in progress.

The action corrects what operators hold about this beacon and brings the passage plan into line with the light in service. It addresses the issue in part. It does not deal with the pending chart correction. The absence of any duty to tell waterway users of a change to an aid is a separate safety issue. No further recommendation has been recorded against this issue.

Clearing of the beacon's sightline deferred unfunded for three works rounds

Work order WO-118, raised in 09/2023, recorded mangrove regrowth on Weerona Islet screening the No. 4 beacon from the northern approach. The trimming it recommended was deferred at the 2023, 2024 and 2025 works rounds as unfunded. By 03/2026 the Authority's register described the sightline as "now fully closed on the northern approach". The islet is 240 m from the beacon, which put it outside the 50 m vegetation scope of schedule 2.4 of the aids to navigation services agreement. The work therefore competed for the general waterways vote rather than the aids to navigation vote.

The issue matters because the aid went on being charted as lit while it could not be seen from the water it served. The deferral very likely contributed to the beacon not being visible from the channel over the half mile before the vessel left it.

Safety action taken

The Harbour Master, Kanooka Heads, funded work order WO-176 from the 2026 waterways contingency. The canopy was trimmed to 2 m on 24/07/2026, restoring the beacon's sightline over the northern approach. A three-yearly trimming cycle has been programmed, and the sightline was re-measured from the inbound track on completion. This action addresses the screening of the No. 4 beacon. It addresses the issue only in part, because the scope that placed the work outside the aids to navigation vote is unchanged.

Recommendation

That [NOT ESTABLISHED: the organisation responsible for the aids to navigation services agreement, to whom this recommendation is addressed] replace the 50 m vegetation scope at schedule 2.4 with a sightline test, so that work needed to keep an aid visible from the water it serves is funded as an aid to navigation, whatever its distance from the structure.

Winter timetable moved the last service into darkness without reopening the route risk assessment

The operator's change process allowed the winter timetable to move the last Weerona Channel service into darkness without reopening the route risk assessment for the channel. That assessment, RRA-07 issue 2, restricted the channel to daylight transits. It required the General Manager Operations' approval and a specific passage plan for any night transit. CHG-2026-014 assessed the move of TF18 to 17:35 as a timetable change only, and the Operations Manager approved it. RRA-07 was not reviewed and no night passage plan was written. The approval of the General Manager Operations, the approver the restriction names, was not sought.

This issue was virtually certain to have existed, and it likely contributed to the occurrence. It led to the channel having no night passage plan, and no hold point if the primary means of fixing was lost.

Its significance extends beyond this occurrence. Between 01/06/2026 and the occurrence, TF18 took a 96-passenger vessel through a 40 m channel in darkness 31 times. Each of those transits contradicted a restriction that was still in force. The process that permitted this asked whether the change was new. It did not ask whether the existing assessment still held for the changed operation.

No safety action has been recorded against this issue.

No night passage plan for the channel, and no hold point if the primary means of fixing is lost

The operator had no night passage plan for the Weerona Channel. Its plan for the channel named no hold point for the loss of the primary means of position fixing. Appendix A to OPS-3 named the

chart plotter as the primary means. It named the visual transit of the No. 4 beacon with the town wharf leading lights as the secondary. It stated that night transits were not scheduled. Nothing in OPS-3 addressed losing the plotter during the transit. The reduced-visibility provisions of OPS-3 3.6 did not apply on the night. It was virtually certain that this gap existed. It was very likely that it contributed to the transit continuing at eight knots after the plotter froze.

The plan was the operator's own defence against this sequence. The master had 114 seconds between the freeze and crossing the channel edge, on flat water with no other traffic. A named hold point would have been available throughout that time. Both masters interviewed said independently that stopping was the right response and that nothing required it.

After the grounding, the General Manager Operations suspended the evening Muntari services. This removed the exposure to darkness transits but provided neither a night plan nor a hold point. It addressed the issue in part.

The investigation recommends that the operator write a night passage plan for the Weerona Channel before any further darkness transit. The plan should name a hold point: the loss of the primary means of position fixing inside the channel requires the way to be taken off the vessel until the position is re-established by another means.

Change-of-master defect brief is verbal, with nothing checking that what is said gets written down

The operator's defect reporting rules allowed a fault on the vessel's primary means of position fixing to pass between masters by word of mouth and go no further. Section 9.3 of OPS-9 required the outgoing master to brief the relieving master on the state of the vessel and any defect. It required nothing of that brief to be written down and nothing of the relieving master afterwards. The chart plotter was not a statutory item on this route, so it fell under section 9.2. That section deferred entry in the defect book to the end of the shift in which a defect was found, which for a fault that comes and goes was no trigger at all. It is virtually certain that this arrangement existed. Both masters independently described the same practice, and the manual bore them out.

The arrangement likely contributed to the grounding. The outgoing master believed the plotter had frozen three or four times in the past year and that none of those freezes had been written up. The firmware fix issued in 11/2024 was never applied. On the night, the master met the fault with no reason to treat it as more than a nuisance.

A contrary reading was also considered. An entry made on the day would not have been reviewed before TF18 sailed. On that reading, the rule is a weakness across the fleet rather than a cause of this grounding. On either reading, intermittent defects in navigation equipment could remain invisible to anyone ashore who was able to act on them.

No safety action has been recorded against this issue.

Forward saloon lifejacket locker not reachable from most of the saloon it serves

The forward saloon lifejacket locker was virtually certain to be out of reach from most of the saloon it served. A survey on 16/07/2026 found the locker face could not be reached from 22 of the 34 forward saloon seats without passing behind the kiosk counter. With people standing at the counter, a 0.6 m inboard gap was the only route. The muster placard directed passengers to a locker they could not see from their seats.

The arrangement approved at build in 2011 had an open bench in that position, with the locker face clear. The counter was fitted in a 2019 interior refit under a minor modification notice. That

notice did not trigger a re-survey of the safety equipment arrangement. The same counter is fitted to the operator's two other 24 m vessels.

It was unlikely that the locker position influenced the grounding or its outcome. The vessel was aground, upright, stable and not sinking. All 38 passengers were in lifejackets within about eight minutes.

The arrangement was nonetheless very likely an important safety issue. In a fire, a night evacuation into the water, or with a deckhand injured, passengers could not reach the lifejackets without crew help. The saloon would wait to be equipped rather than equip itself. The underlying gap was the modification route. An interior refit changed a safety equipment arrangement, and nothing in that route asked whether it had.

No safety action has been recorded against this issue.

Distribution

Title	Name
General Manager Operations	H. Vardon
Safety Manager	P. Nandasiri
Operations Manager	C. Bickerton
Harbour Master, Kanooka Heads	L. Truscott
Aids to Navigation Officer, Kanooka Shire Council	E. Maguire